

STANDARD IEEE 830-1998 / ISO/IEC/IEEE 29148 COMPLIANT

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

GAMING CAFE & ESPORTS ARENA MANAGEMENT SYSTEM

*Real-Time Station Allocation, Live Countdown Billing Engine, Hardware Inventory, Defect Tracking &
Multi-Dimensional PDF Reporting*

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Technology Stack: Java SE (Swing / FlatLaf) & MySQL RDBMS

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1. Introduction

1.1 Purpose of this Document

This Software Requirements Specification (SRS) document details the comprehensive functional, non-functional, behavioral, and architectural requirements for the **Gaming Cafe & Esports Arena Management System**. This document acts as the definitive contract between stakeholders, project evaluators, software architects, and quality assurance personnel. It adheres strictly to the **IEEE Std 830-1998** (Recommended Practice for Software Requirements Specifications) and its modern harmonized successor **ISO/IEC/IEEE 29148:2018**.

1.2 Document Conventions & Prioritization Rules

This specification adheres to standardized typographical and structural conventions:

- **Requirement Identifiers:** Each requirement is tagged with a unique alphanumeric identifier (e.g., FR-AUTH-01, FR-SES-02, NFR-PERF-01) establishing unambiguous traceability.
- **MoSCoW Prioritization Scheme:**
 - **MUST (High):** Absolute mandatory operational requirements. Non-negotiable for system deployment.
 - **SHOULD (Medium):** Highly desirable operational features that enhance efficiency but do not prevent core operation.
 - **COULD (Low):** Optional convenience enhancements implemented if time and resources permit.
- **Database & Code Symbols:** Database tables, column identifiers, classes, and SQL statements are represented in monospaced bold font (e.g., **gaming_sessions.add_minutes**, **pc_table.hourly_rate**).

1.3 Intended Audience & Reading Suggestions

- **Academic Examiners & Evaluators:** Review Section 2 for the rigorous SDLC Waterfall methodology mapping, Section 4 for complete UML diagrams and relational data architecture, and Section 8 for verification test cases.
- **Software Developers & Maintainers:** Focus on Section 4 (Data Dictionaries, SQL relations, and security hashing) and Section 5 (Detailed Functional Specifications).
- **System Administrators & Operations Personnel:** Focus on Section 3 (Operational Environment), Section 5.9 (User Provisioning), and Section 7 (Security and Data Integrity).

1.4 Project Background & Problem Statement

Modern gaming cafes and competitive esports arenas face significant operational bottlenecks when managing dozens of high-performance gaming stations, PlayStation 5 consoles, hourly rental rates, and concurrent customer sessions. The manual, paper-register, or uncoordinated spreadsheet systems traditionally deployed suffer from critical deficiencies:

- **Session Timing Inaccuracies & Revenue Leakage:** Manual recordkeeping makes it difficult to track remaining gameplay time precisely, resulting in unpaid overtime play and financial loss.
- **Cumbersome Mid-Session Extensions:** When gamers request extra time during an ongoing match, manual paper systems fail to update remaining countdowns and total bills dynamically.
- **Untracked Hardware Assets & Defect Dispatches:** Station hardware specifications (CPU, GPU, RAM, Motherboards) and maintenance status lack structured tracking, leading to accidental allocation of damaged PCs.
- **Security & Audit Vulnerabilities:** Physical logbooks lack role-based access control, cryptographic password protection, and tamper-resistant cashier shift audit trails.

The Engineered Solution

The Gaming Cafe Management System replaces fragmented manual processes with an integrated, high-performance desktop platform built on Java Swing, FlatLaf modern UI, and MySQL relational database, delivering real-time countdown tracking, dynamic session extension, cryptographic authentication, and multi-table PDF reporting.

1.5 Project Scope & Operational Boundaries

The system establishes an automated, desktop-based management ecosystem encapsulating the following major capabilities:

- **Secure Role-Based Authentication:** Segregated administrative and operational access using SHA-256 password cryptography, dynamic default password assignment, and mandatory first-time password resets.
- **Interactive Station Grid & KPI Telemetry:** Real-time graphical station cards (PC-01 to PC-08, PS5 consoles) with color-coded status badges (Available = Green, Occupied = Red, Repair = Orange).
- **Session Duration & Real-Time Countdown:** Standard duration presets (1h, 1.5h, 2h, 3h, 4h, 5h, 6h, Custom min 60m) with dynamic hourly rate calculation and per-second remaining countdown timer.
- **Dynamic In-Session Time Extension:** Mid-session '+ Add Time' dialog (+30m, +1h, +2h, Custom) updating the active timer and database records atomically.
- **Hardware Asset Inventory & Maintenance Tracking:** Complete tracking of station specifications (CPU, Motherboard, RAM, GPU, Category, Rate) and defect repair lifecycle.
- **Dedicated Invoicing & PDF Receipts:** Standalone BillFrame window computing cash/card payments, discounts, balance change, and generating iText PDF receipts.
- **Multi-Category PDF Reporting Suite:** Automated export of user activity logs, session history, billing revenue, station inventory, and maintenance logs.

1.6 Applicable Standards & Industry References

1. **IEEE Std 830-1998:** IEEE Recommended Practice for Software Requirements Specifications.
2. **ISO/IEC/IEEE 29148:2018:** Systems and Software Engineering — Life Cycle Processes — Requirements Engineering.
3. **NIST FIPS 180-4:** Secure Hash Standard (SHS) for SHA-256 cryptographic hashing algorithms.
4. **Oracle Java Platform, Standard Edition (Java SE):** Version 21 LTS specifications.
5. **MySQL 8.0 Reference Manual:** Relational Database Management System (RDBMS) ACID transaction compliance.

2. Software Development Life Cycle (SDLC) - Waterfall Model Application

2.1 Justification for the Waterfall SDLC Selection

The classical Waterfall Model was selected as the governance lifecycle methodology for this enterprise gaming cafe management system. Unlike iterative frameworks that introduce continuous scope flux, the gaming cafe domain is characterized by well-established, non-volatile operational rules, rigid hardware relationships, and clear sequential dependencies:

- **Predictable, Well-Understood Problem Domain:** Gaming cafe workflows (station selection → duration allocation → gameplay countdown → session termination → billing settlement) are linear and deterministic.
- **Rigid Relational Dependencies:** Relational database schemas (foreign keys linking user, user_logs, pc_table, gaming_sessions, pc_maintenance, and invoices) must be fully modeled and normalized to Third Normal Form (3NF) before presentation forms or business validation logic can be written.
- **Quality Assurance & Verification Rigor:** Real-time session timing and financial billing calculation require formal algorithm verification before deployment, matching Waterfall's distinct phase-gate milestones.

2.2 Detailed Waterfall Phase Breakdown

Phase 1: Requirements Analysis & Elicitation: Extensive domain analysis and stakeholder interviews were conducted with gaming cafe owners, counter staff, and tournament coordinators. Operational pain points were extracted and formalized into this IEEE 830 compliant SRS document.

Phase 2: Architectural & Detailed System Design: The architectural blueprint decomposed the application into a 3-Tier Layered Architecture (Presentation, Business Logic, and Data Access). Detailed database schemas with strict primary and foreign key constraints were designed. Form blueprints were structured in NetBeans GUI Builder XML (.form).

Phase 3: Implementation, Coding & Security Engineering: Physical development took place in Java utilizing NetBeans IDE and Ant Build Tooling. Clean coding standards were enforced with top-level imports only, strict exception logging, SHA-256 one-way hashing, FlatLaf modern UI integration, and parameterized JDBC PreparedStatements.

Phase 4: Integration, Verification & System Testing: Individual modules were integrated and subjected to unit testing (NIC regex, phone numbers, duration bounds), integration testing (timer synchronization, add-time DB updates), and security audits.

Phase 5: Deployment, Configuration & UAT: The software was packaged into an executable JAR archive with all bundled dependencies (flatlaf, mysql-connector-j, itextpdf). The MySQL schema was initialized, and User Acceptance Testing (UAT) was executed with counter staff.

Phase 6: Maintenance, Support & Operational Evolution: Corrective maintenance (bug fixing), adaptive maintenance (updating hardware tariffs or OS scaling), and perfective maintenance (cloud multi-branch sync and QR token integration).

2.3 Waterfall Milestones, Deliverables & Quality Gates Table

Waterfall Phase	Inputs	Key Engineering Activities	Outputs / Deliverables	Quality Review Gate
Phase 1: Requirements	Stakeholder interviews, legacy paper logs	Problem analysis, constraint elicitation, operational modeling, MoSCoW prioritization	Baselined SRS Document (IEEE 830), Feasibility Report	Requirements Review Milestone (SRS Sign-off)
Phase 2: Design	Baselined SRS Document	3-Tier architectural partitioning, ERD normalization (3NF), GUI wireframing	System Design Document (SDD), Database DDL Scripts, UI Form Blueprints	Preliminary & Critical Design Reviews (PDR/CDR)
Phase 3: Coding	SDD, DB DDL Scripts, UI Wireframes	Java Swing development, FlatLaf styling, JDBC integration, SHA-256 crypto implementation	Compiled Source Code (.java), Form Definitions (.form), Ant Build Scripts	Code Quality & Peer Inspection Gate
Phase 4: Testing	Compiled System Modules, Test Plan	Unit testing, integration testing, timer countdown stress tests, regression suites	Test Case Results, Traceability Matrix, Bug Resolution Log	Test Readiness & Verification Gate
Phase 5: Deployment	Verified Build, Production DB Schema	JAR packaging, MySQL database installation, admin credential provisioning, UAT execution	Deployable GamingCafe.jar, User Manual, UAT Sign-Off Certificate	Operational Acceptance Milestone
Phase 6: Maintenance	User defect tickets, hardware changes	Log monitoring, bug fixing, schema adaptation, performance tuning	Service Patches, Version Updates, Maintenance Logs	Post-Implementation Review Gate

3. Overall System Description

3.1 Product Perspective & Context

The Gaming Cafe Management System is a standalone desktop application communicating with a high-performance relational database management system (MySQL) via the Java Database Connectivity (JDBC) API. It replaces fragmented paper registers and unlinked spreadsheets with an authoritative, unified transactional store. The application runs natively on client workstations deployed at the reception, dispatch, and administrative management desks.

3.2 3-Tier Layered Architecture Pattern

To ensure high cohesion, low coupling, maintainability, and clean separation of concerns, the system is engineered around a 3-Tier Layered Architecture:

- **Presentation Layer (GUI Tier):** Built with Java Swing and styled with the FlatLaf modern UI library. Manages user visual interactions, responsive card layout transitions (CardLayout), dynamic station cards, and event listeners.
- **Business Logic Layer (Controller Tier):** Implements domain rules, real-time session countdown timers (javax.swing.Timer), duration and price calculations, input validation algorithms (regex for NIC, phone numbers), cryptographic password digestion (SHA-256), and role-based feature gating.
- **Data Access Layer (Persistence Tier):** Encapsulated by centralized database connection management (db.java) and parameterized JDBC PreparedStatement objects. Guarantees ACID transactional compliance, isolates the application from underlying database specifics, and protects against SQL injection.

3-TIER LAYERED ARCHITECTURE SPECIFICATION

Presentation, Business Logic Controller, and JDBC Data Access Architecture

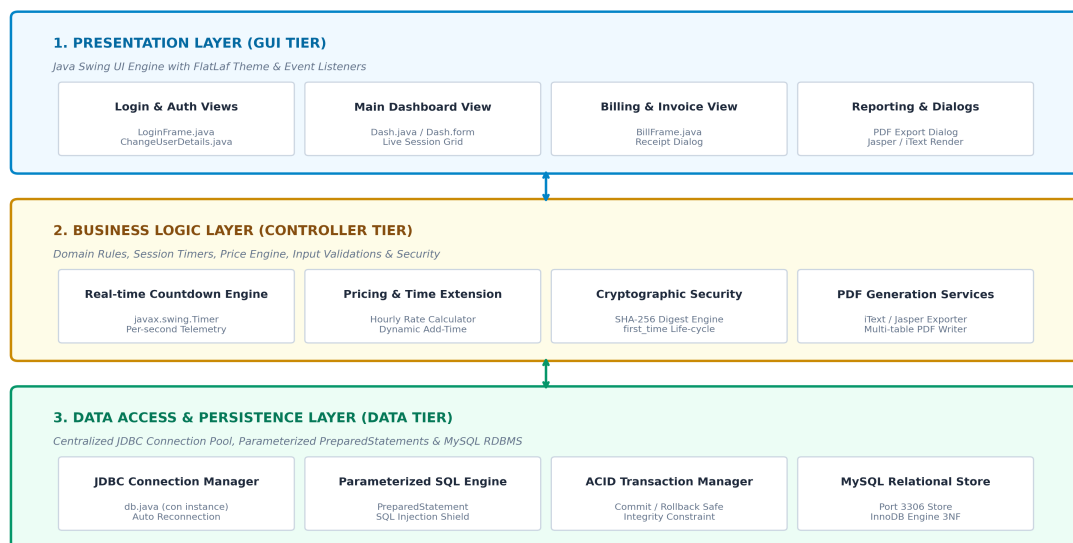


Figure 3.1: 3-Tier Layered Architecture Pattern of the Gaming Cafe Platform

3.3 User Classes, Personas & Operational Responsibilities

User Role	Persona Profile	System Capabilities & Access Rights	Operational Security Level
System Administrator	Cafe Owner / General Manager / IT Administrator	Full system privileges. Can access Dashboard, Live Sessions, Station Inventory Management, Maintenance, History, multi-category PDF Reporting, and exclusive User Administration (create staff/admin, reset user flags, toggle active/inactive).	Tier 1 - Full Unrestricted Administrative Access
Cafe Staff / Cashier	Front-Desk Receptionist / Counter Cashier	Operational access. Can start sessions, allocate duration, execute dynamic '+ Add Time' extensions, end sessions, generate BillFrame invoices, collect payments, and log repair requests. Blocked from User Administration and system settings.	Tier 2 - Operational Business Access Only
Gamer / Customer	Enrolled Player / Esports Competitor	Indirect user. Registration details, contact phone, allocated gaming time, and payment records are tracked. Can utilize Token/QR pass for instant station check-in.	Tier 3 - Data Entity / Customer Resource Tier

3.4 Operational Environment & System Requirements

Client Workstation Hardware Specifications:

- **Processor:** Intel Core i3 (2.0 GHz) or AMD equivalent (Intel Core i5 recommended).
- **Random Access Memory (RAM):** Minimum 4 GB RAM (8 GB recommended for optimal JVM performance).
- **Hard Disk Storage:** Minimum 500 MB free storage for application runtime and local PDF logging.
- **Display Resolution:** Minimum 1366 × 768 pixels (Full HD 1920 × 1080 recommended).
- **Input Devices:** Standard QWERTY keyboard, mouse, and optional 2D barcode/QR scanner.

Software & Platform Environment:

- **Operating System:** Microsoft Windows 10 (64-bit) or Windows 11. Cross-compatible with Linux (Ubuntu 20.04+) and macOS (12+).
- **Java Runtime Environment:** Oracle OpenJDK or Eclipse Temurin JRE 8, 11, 17, or 21 (LTS).
- **Database Management System:** MySQL Community Server 8.0 or 8.4 LTS running on port 3306.
- **JDBC Database Driver:** MySQL Connector/J 8.0.33+ included in application distribution.
- **UI Skinning Engine:** FlatLaf Modern Look and Feel Version 3.5.4+.
- **PDF Generation Engine:** iText PDF Library 5.5+ / JasperReports.

3.5 Design & Implementation Constraints

1. **Desktop-First Native Architecture:** Developed as a high-responsiveness native Java desktop GUI to operate without dependencies on external cloud web servers or continuous internet connectivity.
2. **Irreversible Password Security:** Passwords must never be stored in plain text. Passwords must be hashed using one-way SHA-256 with Base64 encoding.
3. **Relational Integrity:** Strict Foreign Key cascades must be implemented so that deleting or updating records maintains referential integrity across user, user_logs, pc_table, gaming_sessions, pc_maintenance, and invoices.
4. **GUI Thread Safety:** Long-running database operations and countdown timer ticks must update Swing components strictly within the Event Dispatch Thread (`java.awt.EventQueue.invokeLater`).

4. System Architecture, Unified Modeling Language (UML) & Data Models

4.1 Use Case Diagram & Actor Responsibilities

The behavioral interactions between the primary actors (System Administrator, Cafe Staff / Cashier) and the system boundary are modeled below. System Administrator inherits all staff operational use cases with exclusive rights over user provisioning and global report compilation.

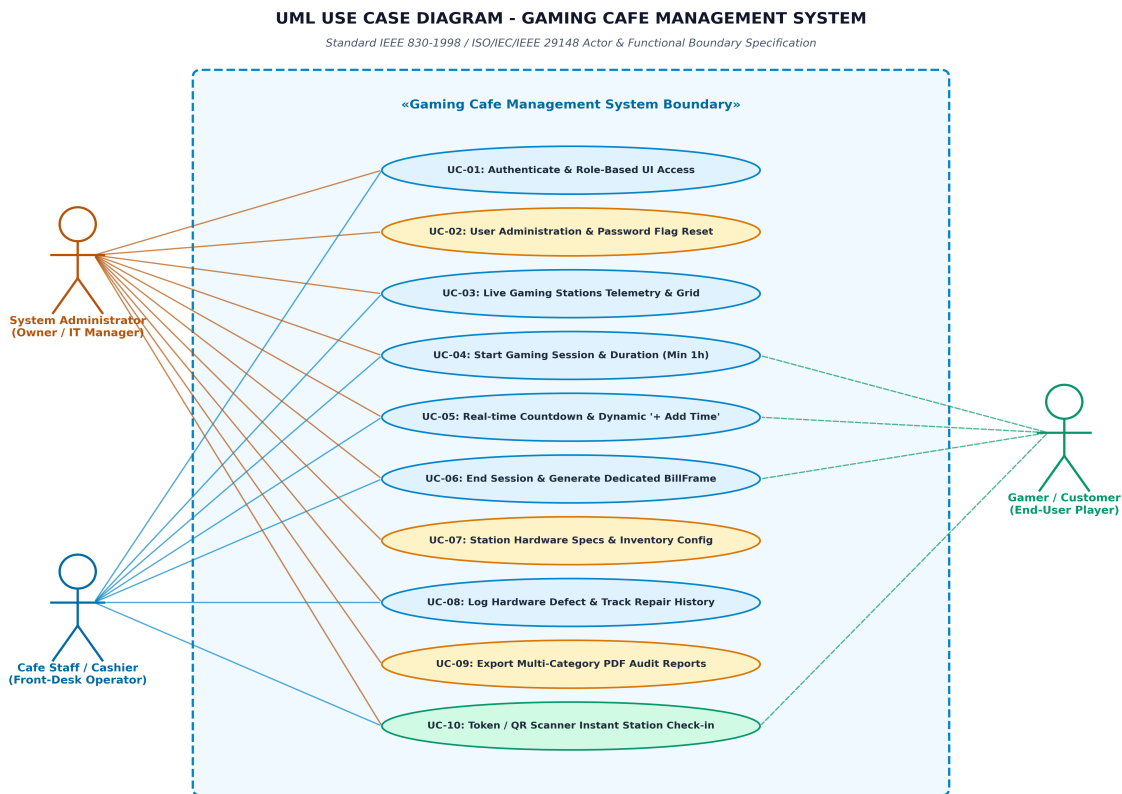


Figure 4.1: UML Use Case Diagram for Gaming Cafe Management System

4.2 Class Diagram & Object-Oriented Domain Model

The static structure of the Java software architecture is depicted in the Class Diagram below, showing domain entities (User, GamingStation, GamingSession, Invoice, PCMaintenance, UserLogs), utility classes (DBConnection), and GUI controllers (Dash, BillFrame).

UML CLASS DIAGRAM - OBJECT-ORIENTED SYSTEM DOMAIN MODEL

Java Swing Controllers, Domain Entities, Data Access Objects (DAO) & Persistence Classes

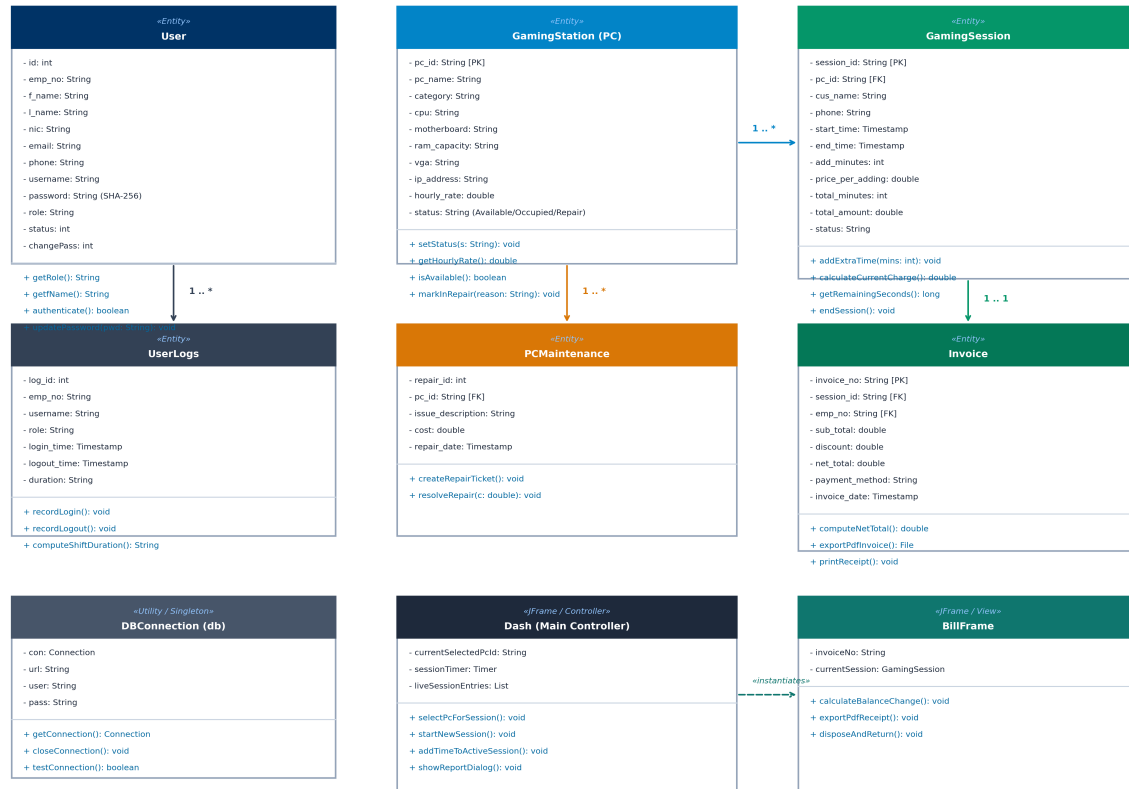


Figure 4.2: UML Class Diagram (Domain Model, Entities & Controller Classes)

4.3 Sequence Diagram: Session Booking, Live Countdown & Billing Lifecycle

The dynamic runtime message interactions between Cashier, Dash Controller, SessionTimer, MySQL Database, and BillFrame during a full gaming session lifecycle are modeled below:

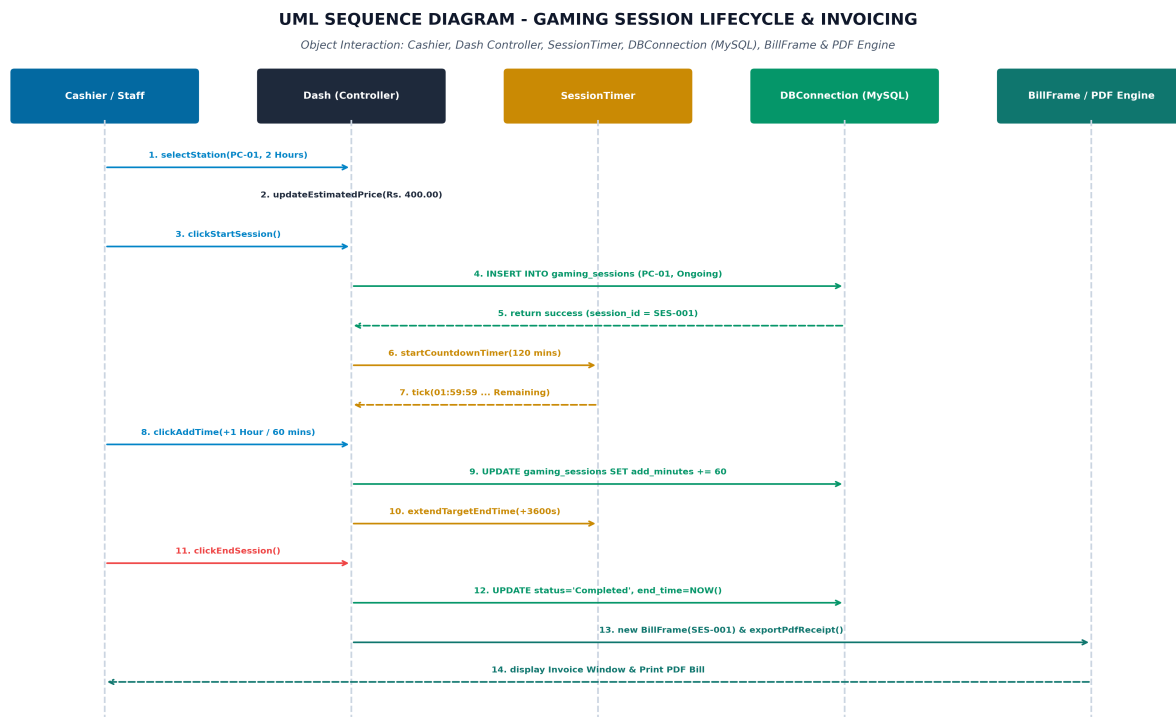
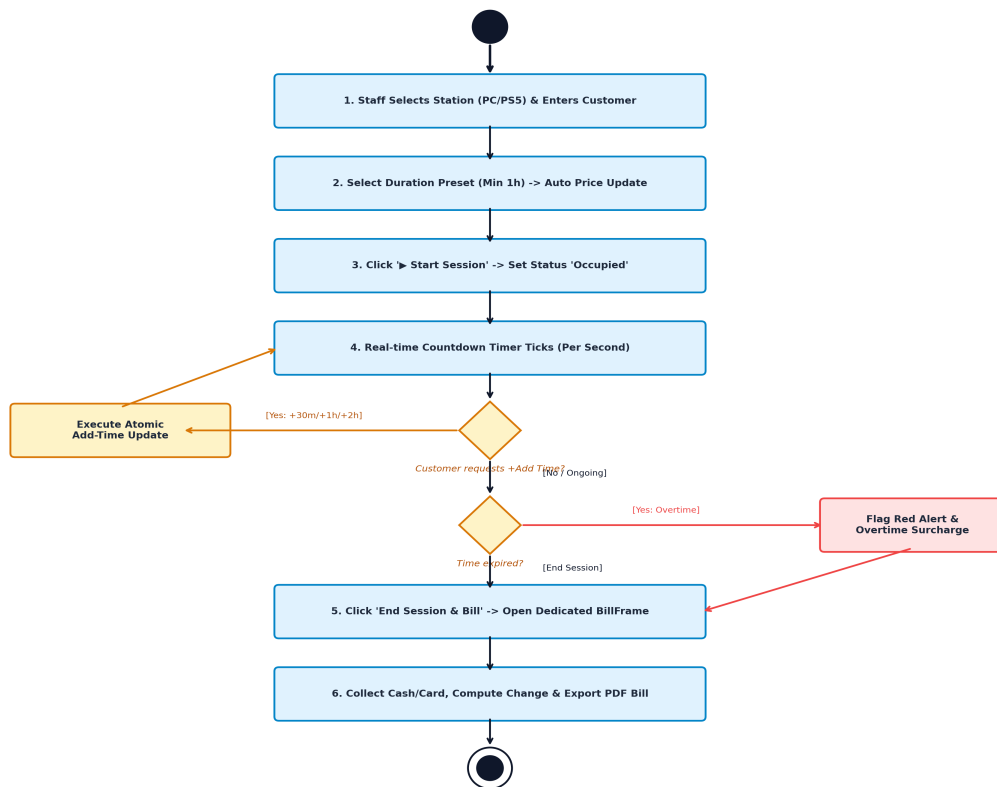


Figure 4.3: UML Sequence Diagram - Gaming Session Allocation, Dynamic Time Extension & Invoicing

4.4 Activity Diagram: Operational Control Flow & Decision Logic

The procedural workflow of session dispatching, duration selection, real-time countdown execution, dynamic mid-session time extensions (+30m, +1h, +2h, Custom), overtime handling, and PDF invoice generation is illustrated in the Activity Diagram below:

UML ACTIVITY DIAGRAM - SESSION DISPATCH & BILLING WORKFLOW*Operational Control Flow with Decision Diamonds, Dynamic Extension Loop & Overtime Logic**Figure 4.4: UML Activity Diagram - Station Dispatch, Extension Loop & Billing Flowchart*

4.5 State Machine Diagram: Station Finite State Transitions

Each gaming station (PC or PS5) operates as a finite state-machine with three distinct operational states: Available (Green), Occupied (Red), and Under Maintenance / Repair (Orange). State transitions are triggered by cashier actions and technician repair resolutions:

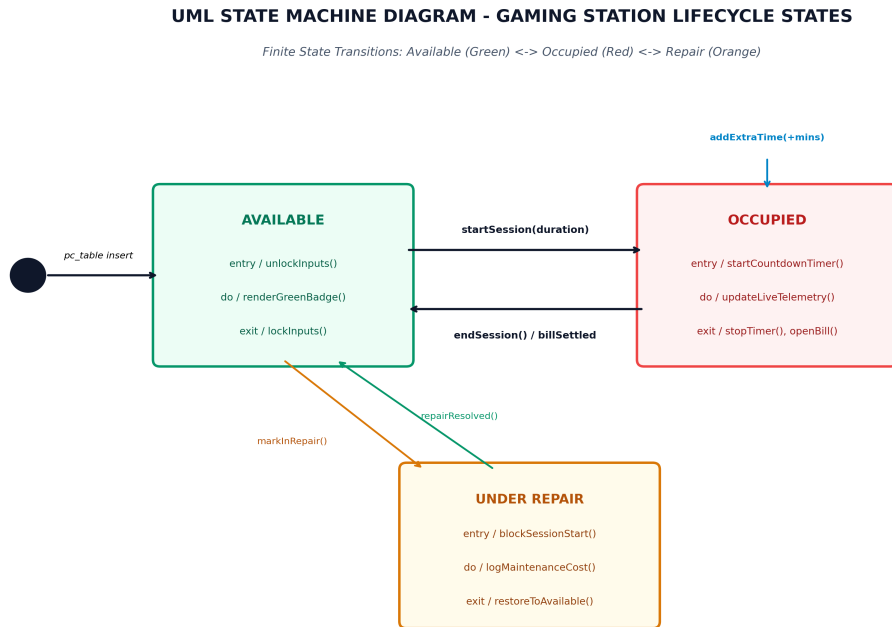


Figure 4.5: UML State Machine Diagram - Gaming Station Finite State Lifecycle

4.6 Relational Database Schema & Data Dictionaries

The relational database schema is normalized to Third Normal Form (3NF), eliminating transitive and partial dependencies while enforcing domain constraints. Below is the exact production database schema diagram corresponding to the MySQL database instance:

EXACT RELATIONAL DATABASE SCHEMA & ENTITY RELATIONSHIPS

MySQL Database: 'gaming_cafe' Normalized Schema as Deployed in Production Environment

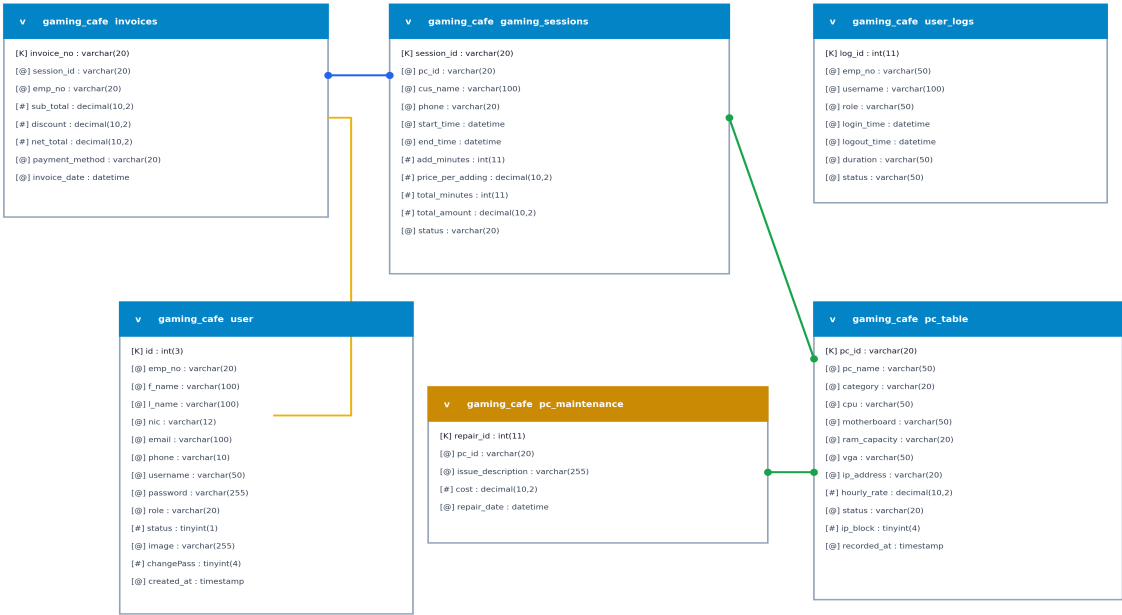


Figure 4.6: Exact Relational Database Schema & Foreign Key Relationships

Data Dictionaries (Production MySQL Tables)

Table 1: user (User Accounts & Authentication Credentials)

Column Name	Data Type	Constraints	Default	Description & Validation Rules
id	int(3)	PRIMARY KEY, AUTO_INC	None	Unique auto-generated identifier for the user account.
emp_no	varchar(20)	NOT NULL, UNIQUE	None	Employee registration number (e.g., 'EMP-001').
f_name	varchar(100)	NOT NULL	None	First name of the staff member.
l_name	varchar(100)	NOT NULL	None	Last name / surname of the staff member.
nic	varchar(12)	NOT NULL, UNIQUE	None	National Identity Card number (validated via regex).
email	varchar(100)	NULL	None	Email address for notifications.
phone	varchar(10)	NOT NULL	None	Mobile telephone contact number (10 digits).
username	varchar(50)	NOT NULL, UNIQUE	None	Unique login username (3-50 characters).
password	varchar(255)	NOT NULL	None	SHA-256 cryptographic one-way digest encoded in Base64.
role	varchar(20)	NOT NULL	'Staff'	Access permission role: 'Admin' or 'Staff'.
status	tinyint(1)	NOT NULL	1	Account operational state: 1 = Active, 0 = Inactive.
image	varchar(255)	NULL	None	Relative file path to user profile avatar.
changePass	tinyint(4)	NOT NULL	1	Flag for mandatory first-time password reset: 1 = Required, 0 = Active.
created_at	timestamp	DEFAULT CURRENT	Current	Timestamp of account registration.

Table 2: user_logs (Staff Authentication & Shift Audit Trail)

Column Name	Data Type	Constraints	Default	Description & Validation Rules
log_id	int(11)	PRIMARY KEY, AUTO_INC	None	Unique audit log record serial number.
emp_no	varchar(50)	FOREIGN KEY → user	None	Referenced employee number of authenticated user.
username	varchar(100)	NOT NULL	None	Captured username at login timestamp.
role	varchar(50)	NOT NULL	None	User role at authentication (Admin / Staff).
login_time	datetime	NOT NULL	CURRENT	Timestamp of successful login.
logout_time	datetime	NULL	None	Timestamp of clean logout session termination.
duration	varchar(50)	NULL	None	Calculated active shift duration (e.g., '04h 25m 10s').
status	varchar(50)	NOT NULL	'Active'	Session state: 'Active', 'Logged Out'.

Table 3: pc_table (Gaming Stations & Hardware Inventory)

Column Name	Data Type	Constraints	Default	Description & Validation Rules
pc_id	varchar(20)	PRIMARY KEY	None	Unique station identifier (e.g., 'PC-01', 'PS5-02').
pc_name	varchar(50)	NOT NULL	None	Descriptive station alias (e.g., 'RTX 4070 Station').
category	varchar(20)	NOT NULL	'PC'	Hardware category: 'PC', 'PS5', 'Racing Sim'.
cpu	varchar(50)	NOT NULL	None	Central Processing Unit model.
motherboard	varchar(50)	NOT NULL	None	Motherboard chipset / model.
ram_capacity	varchar(20)	NOT NULL	None	Installed RAM capacity & speed (e.g., '32GB DDR5').
vga	varchar(50)	NOT NULL	None	Dedicated Graphics Processing Unit (e.g., 'RTX 4070 12GB').
ip_address	varchar(20)	NULL	None	LAN IP address for remote station management.
hourly_rate	decimal(10,2)	NOT NULL	200.00	Base hourly rental tariff in LKR.
status	varchar(20)	NOT NULL	'Available'	Operational status: 'Available', 'Occupied', 'Repair'.
ip_block	tinyint(4)	NOT NULL	0	Network access block status (0 = Allowed, 1 = Blocked).
recorded_at	timestamp	DEFAULT CURRENT	Current	Timestamp when station was registered.

Table 4: gaming_sessions (Live & Historic Play Sessions)

Column Name	Data Type	Constraints	Default	Description & Validation Rules
session_id	varchar(20)	PRIMARY KEY	None	Unique session identifier (e.g., 'SES-001').
pc_id	varchar(20)	FOREIGN KEY → pc_table	None	Referenced gaming station allocated for play.
cus_name	varchar(100)	NOT NULL	'Walk-in'	Gamer / Player customer name.
phone	varchar(20)	NULL	None	Mobile telephone contact number of customer.
start_time	datetime	NOT NULL	CURRENT	Timestamp when session gameplay commenced.
end_time	datetime	NULL	None	Timestamp when session was terminated.
add_minutes	int(11)	NOT NULL	0	Cumulative extra time added during session in minutes.
price_per_adding	decimal(10,2)	NOT NULL	0.00	Additional charge incurred via '+ Add Time'.
total_minutes	int(11)	NOT NULL	60	Total allocated playing duration (Initial + Add Time).
total_amount	decimal(10,2)	NOT NULL	0.00	Total calculated session charge before discounts.
status	varchar(20)	NOT NULL	'Ongoing'	Session state: 'Ongoing', 'Completed', 'Cancelled'.

Table 5: pc_maintenance (Defect Logging & Repair Records)

Column Name	Data Type	Constraints	Default	Description & Validation Rules
repair_id	int(11)	PRIMARY KEY, AUTO_INC	None	Unique defect repair ticket serial number.
pc_id	varchar(20)	FOREIGN KEY → pc_table	None	Station currently under hardware repair.
issue_description	varchar(255)	NOT NULL	None	Reported hardware failure description.
cost	decimal(10,2)	NOT NULL	0.00	Total maintenance / replacement cost incurred in LKR.
repair_date	datetime	NOT NULL	CURRENT	Timestamp when repair was logged.

Table 6: invoices (Financial Invoicing & Payment Records)

Column Name	Data Type	Constraints	Default	Description & Validation Rules
invoice_no	varchar(20)	PRIMARY KEY	None	Unique invoice serial number (e.g., 'INV-0104').
session_id	varchar(20)	FOREIGN KEY → gaming_sessions	None	Referenced session record billed.
emp_no	varchar(20)	FOREIGN KEY → user	None	Employee / Cashier who finalized transaction.
sub_total	decimal(10,2)	NOT NULL	0.00	Subtotal amount before promotional discounts.
discount	decimal(10,2)	NOT NULL	0.00	Promotional discount deducted from bill.
net_total	decimal(10,2)	NOT NULL	0.00	Final net amount payable by customer.
payment_method	varchar(20)	NOT NULL	'Cash'	Payment method utilized: 'Cash', 'Card', 'QR Token'.
invoice_date	datetime	DEFAULT CURRENT	Current	Timestamp of invoice settlement.

4.7 Real-Time Session Countdown & Dynamic Add-Time Algorithms

The system executes a real-time countdown algorithm updating per-second timestamps via **javax.swing.Timer**. When a player requests additional time (+30m, +1h, +2h, Custom), the system executes an atomic parameterized transaction:

```
-- Atomic Time Extension Query Executed via PreparedStatement:
UPDATE gaming_sessions
SET add_minutes = add_minutes + ?,
    total_minutes = total_minutes + ?,
    price_per_adding = ?,
    total_amount = total_amount + ?
WHERE session_id = ? AND status = 'Ongoing';
```

4.8 Cryptographic Security Architecture & Password Lifecycle

User credentials are protected using NIST FIPS 180-4 compliant SHA-256 one-way hashing. If `changePass == 1`, the user is intercepted upon login and redirected to `ChangeUserDetails.java` before dashboard access is permitted. Administrators can reset any user's credentials back to the default password via the `btnResetUser` action button in User Administration.

5. Specific Functional Requirements (Modules 1 to 9)

5.1 Module 1: Authentication & Access Control

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-AUTH-01	Credential Hashing Authentication	MUST	The system shall authenticate usernames/NIC and passwords by digesting input via SHA-256 and matching digests against the user table.
FR-AUTH-02	First-Time Login Interception	MUST	If a logging-in user has changePass = 1, the system shall intercept navigation and display ChangeUserDetails.java.
FR-AUTH-03	Mandatory Password Reset	MUST	ChangeUserDetails form shall require entering new password and confirmation, validate minimum 4 characters, hash with SHA-256, set changePass = 0, and advance to Dashboard.
FR-AUTH-04	Role-Based UI Gating	MUST	Upon login, the system shall evaluate user role. If role is 'Staff', administrative navigation buttons (btnAdmin, User Management, Global Settings) shall be hidden.
FR-AUTH-05	User Activity Shift Logging	MUST	Upon every successful authentication, the system shall insert a record into user_logs capturing emp_no, username, role, and login_time.
FR-AUTH-06	Secure Session Logout	MUST	The system shall allow users to log out, updating logout_time and duration in user_logs, terminating active dashboard windows, and returning to LoginFrame.

5.2 Module 2: Central Operational Dashboard & Real-Time KPI Telemetry

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-DASH-01	Live Station KPI Telemetry	MUST	The dashboard shall execute aggregation queries to compute and render Total Stations, Active Playing Sessions, Available PCs, and Stations in Repair in real-time summary cards.
FR-DASH-02	Interactive Station Grid Rendering	MUST	The dashboard shall dynamically render interactive cards for all stations (PC-01 to PC-08, PS5 consoles) with color-coded status badges (Available = Green, Occupied = Red, Repair = Orange).
FR-DASH-03	One-Click Station Selection	MUST	Clicking any station card in the grid shall immediately populate the Session Control side-panel with station name, category, hourly rate, and active session telemetry.
FR-DASH-04	Digital Clock & Date Header	MUST	The system shall display live synchronized system date and digital time in the top application header.
FR-DASH-05	Collapsible Hamburger Navigation	SHOULD	The system shall support toggling the sidebar navigation menu via btnHam to maximize workspace area on compact displays.

5.3 Module 3: Live Gaming Sessions & Active Station Monitoring

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-LIVE-01	Live Sessions Tabular View	MUST	The system shall provide a dedicated Live Sessions table (JTableLive) displaying all active ongoing sessions with Customer Name, Phone, Station, Start Time, Duration, and Live Timer.
FR-LIVE-02	Real-time Live Table Tick Refresh	MUST	The Live Sessions table shall update every second via javax.swing.Timer to show synchronized countdown timestamps across all concurrent players.
FR-LIVE-03	Live Search & Customer Filtering	MUST	The system shall filter the live sessions table dynamically as characters are typed into txtLiveSearch by customer name, phone number, or PC ID.
FR-LIVE-04	One-Click Session End from Live Table	MUST	Double-clicking or selecting an active row in the live table shall allow instant session termination and invoice generation.
FR-LIVE-05	Overtime Visual Highlighting	MUST	Active sessions exceeding their allocated duration shall be highlighted in high-contrast red font with overtime surcharge notification.

5.4 Module 4: Session Duration Allocation & Dynamic Time Extension

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-SES-01	Standard Duration Presets	MUST	The system shall provide a duration selection ComboBox (cmbSessionDuration) with options: 1h (60m), 1.5h (90m), 2h (120m), 3h (180m), 4h (240m), 5h (300m), 6h (360m), and Custom Minutes.
FR-SES-02	Minimum 1-Hour Duration Validation	MUST	The system shall enforce a minimum session duration of 60 minutes (1 Hour) for both standard presets and custom minute entries.
FR-SES-03	Real-time Estimated Charge Calculation	MUST	Selecting any duration preset shall instantly calculate and render the estimated session price: (Allocated_Minutes / 60.0) * Station_Hourly_Rate.
FR-SES-04	Dynamic In-Session '+ Add Time'	MUST	The system shall provide an '■ + Add Time' button during active sessions allowing staff to add +30m, +1h, +2h, or Custom minutes, updating the active countdown and database record immediately.
FR-SES-05	Session Start Validation & Dispatch	MUST	Clicking '■ Start Session' shall validate customer name, verify station availability, insert record into gaming_sessions, set station status to 'Occupied', and start the countdown timer.

5.5 Module 5: Station & Hardware Inventory Management

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-PC-01	Hardware Specification Logging	MUST	The system shall record station hardware specifications: Station ID, Name, Category (PC/PS5), Hourly Rate, CPU, Motherboard, RAM, and Dedicated Graphics Card (VGA).
FR-PC-02	Station Inventory Tabular View	MUST	The system shall display all registered stations in a styled table with real-time status badges and hardware component summaries.
FR-PC-03	Hardware Specification Editing	MUST	Authorized administrators shall be able to update station hardware specifications, component upgrades, and hourly rental tariffs.
FR-PC-04	Station Decommissioning & Removal	MUST	The system shall permit removing retired stations upon administrative confirmation while preserving historic session logs.
FR-PC-05	Hardware Category Filtering	SHOULD	The system shall support filtering stations by hardware category (PC High-Tier, PC Mid-Tier, PS5 Console) for rapid dispatch.

5.6 Module 6: Hardware Maintenance, Defect Logging & Repair History

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-REP-01	Station Defect Dispatch	MUST	Staff shall be able to mark any malfunctioning station as 'Repair', entering a detailed defect reason (e.g., 'GPU Artifacts', 'RAM Failure') and taking the station off-line.
FR-REP-02	Active Repairs Management Table	MUST	The system shall maintain an active repair table (jTable2) listing all stations currently under maintenance with reported issues and dates.
FR-REP-03	Repair Resolution & Return to Service	MUST	Upon technician completion, staff shall log the repair cost, restoring station status to 'Available' in pc_table and archiving the record to pc_maintenance.
FR-REP-04	Maintenance History & Cost Auditing	MUST	The system shall display historical maintenance records in pc_maintenance table to audit total hardware repair expenditures over time.
FR-REP-05	Maintenance PDF Report Export	SHOULD	Administrators shall be able to compile and export the complete hardware maintenance and repair expenditure log as a formatted PDF document.

5.7 Module 7: Invoicing, Billing Engine & PDF Receipt Export

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-BILL-01	Dedicated BillFrame Presentation	MUST	Clicking '■ End Session & Bill' shall terminate the active session, calculate total charges, and open a dedicated, modern BillFrame window.
FR-BILL-02	Itemized Invoice Breakdown	MUST	BillFrame shall display invoice serial number, station name, category, customer name, phone, start/end timestamps, total duration, hourly rate, and subtotal amount.
FR-BILL-03	Promotional Discount Calculation	MUST	BillFrame shall support applying special promotional discounts, recalculating Net Payable Amount in real-time.
FR-BILL-04	Cash & Change Due Calculator	MUST	Entering cash received from customer shall automatically compute and display the change due (Balance = Cash_Received - Net_Amount).
FR-BILL-05	Payment Method Selection	MUST	Staff shall record the payment method (Cash / Card / QR Token) before finalizing the invoice transaction.
FR-BILL-06	iText PDF Receipt Export & Print	MUST	Clicking 'Print / Save PDF Bill' shall compile and export an official PDF invoice receipt and prompt the user to immediately view/print the document.

5.8 Module 8: Business Intelligence & Multi-Category PDF Reporting

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-REP-01	Multi-Category Report Generation Dialog	MUST	The system shall provide an interactive option dialog (showReportGenerationDialog) allowing administrators to generate 5 distinct PDF reports.
FR-REP-02	User Activity Logs & Attendance Report	MUST	The system shall compile and export all staff login, logout, and shift duration records into a structured 'User Activity Logs Report' PDF.
FR-REP-03	Session History & Usage Utilization Report	MUST	The system shall compile and export all historic gaming sessions into a structured 'PC Session History Report' PDF.
FR-REP-04	Invoices & Revenue Billing Report	MUST	The system shall compile and export all finalized customer invoices, payment methods, discounts, and total earnings into an 'Invoices Billing Report' PDF.
FR-REP-05	Station Hardware Inventory Report	MUST	The system shall compile and export all registered stations, hardware specs, and hourly rates into a 'PC Inventory Report' PDF.
FR-REP-06	Universal PDF Exporter Engine	MUST	The exportTableToPdf engine shall automatically extract JTable column models, apply high-contrast header styling, alternate row colors, and save files to user home directory.

5.9 Module 9: User Administration & Security Provisioning

Req ID	Requirement Title	Priority	Detailed Functional Specification
FR-USR-01	Staff Account Provisioning	MUST	Administrators shall create new staff accounts specifying First Name, Last Name, NIC, Phone, Email, Username, Role (Staff/Admin), and Password.
FR-USR-02	Automatic Default Password Assignment	MUST	If password fields are left blank during account provisioning, the system shall assign the default password ('1234') hashed via SHA-256 with changePass = 1.
FR-USR-03	Administrative Credential Reset	MUST	The system shall provide a 'Reset User' action button (btnResetUser) allowing administrators to restore any user's credentials to the default password and reset changePass flag.
FR-USR-04	Account Inactivation & Activation	MUST	Administrators shall be able to toggle staff account operational states between Active (1) and Inactive (0) to prevent unauthorized shift logins.
FR-USR-05	Dynamic Action Button Visibility	MUST	Action buttons (btnUpdateUser, btnDeleteUser, btnClearUser, btnResetUser) shall remain hidden until a specific record is selected in tableUser.
FR-USR-06	Self-Deletion Prevention	MUST	The system shall block administrators from deleting or inactivating their own currently authenticated user account, preventing system lockout.

6. External Interface Requirements

6.1 Graphical User Interfaces (GUI) & Dynamic Controls

The visual presentation is engineered natively in Java Swing with FlatLaf vector styling. Below are the operational blueprints of the main system interfaces:

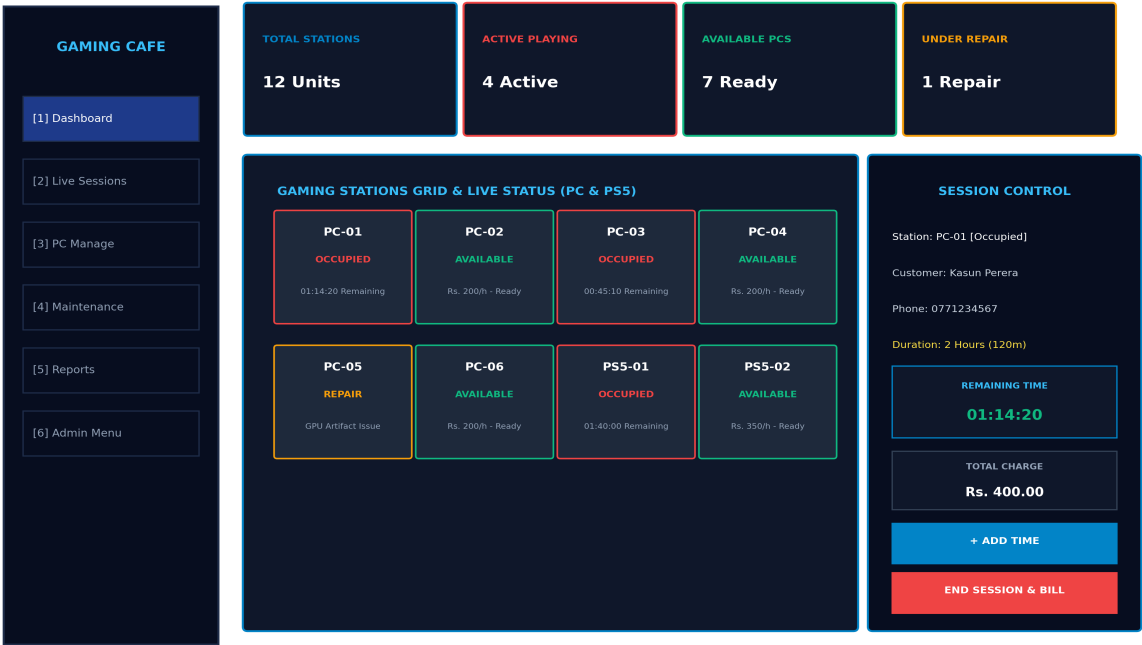


Figure 6.1: Central Operational Dashboard with Live Stations Grid, KPI Cards, and Session Control Panel

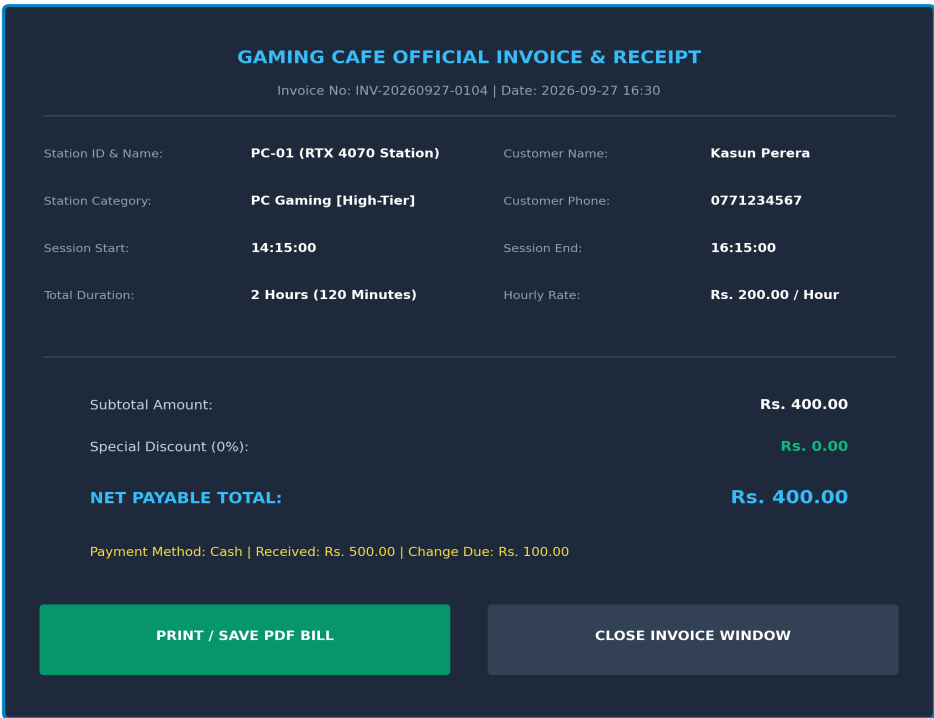


Figure 6.2: Dedicated Invoicing & Bill Receipt Generation Window (BillFrame)



Figure 6.3: Multi-Dimensional PDF Business Intelligence & Audit Reporting Dialog

6.2 Hardware Interfaces & Token / QR Code Instant Session Check-In Scanner

The system is engineered to interface seamlessly with standard USB / Bluetooth 2D barcode and QR code scanners acting as Human Interface Devices (HID keyboard wedge). When a customer presents an official membership token or QR pass, the scanner transmits the player token string into the active station dispatch listener, triggering automated station unlocking and session initialization:

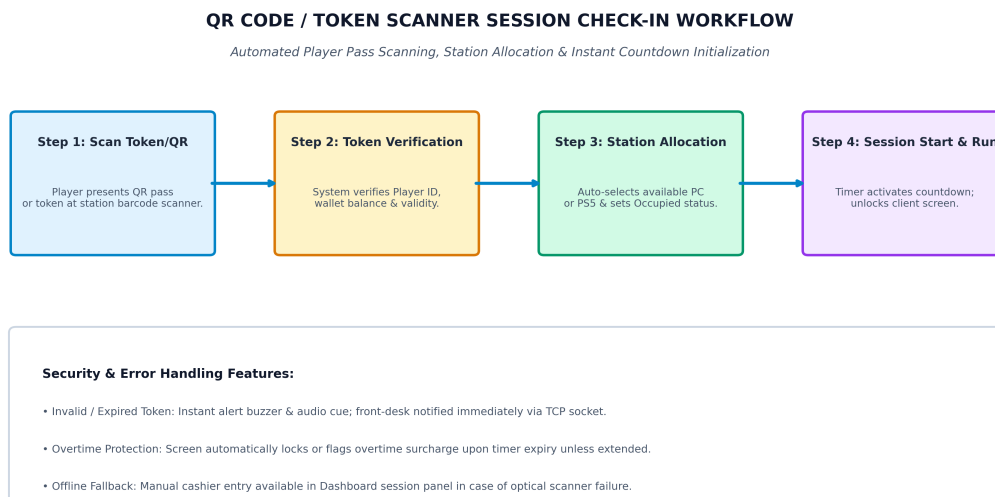


Figure 6.4: Token & QR Scanner Automated Session Check-In and Station Allocation Workflow

6.3 Software & Driver Interfaces

- **Java Database Connectivity (JDBC):** `com.mysql.cj.jdbc.Driver` establishes connection pooling with the MySQL database instance on port 3306.
- **FlatLaf Look-and-Feel Engine:** `com.formdev.flatlaf.FlatIntelliJLaf` provides high-DPI scaling and vector styling.
- **iText PDF Library / JasperReports:** Generates official PDF receipts and multi-table business audit reports.

6.4 Communications & Network Protocols

The system communicates with the MySQL database instance over TCP/IP socket connections (default port 3306). Localhost loopback sockets (127.0.0.1) are utilized for single-workstation standalone deployments, while standard Local Area Network (LAN) intranet IP addresses (e.g., 192.168.1.100) are fully supported for multi-counter cafe setups.

7. Non-Functional Requirements (NFR)

7.1 Performance & Scalability Benchmarks

NFR ID	Quality Characteristic	Measurable Target / Benchmark
NFR-PERF-01	Query Response Time	Station grid and session data queries shall execute and render in < 250 ms under standard database loads ($\leq 50,000$ records).
NFR-PERF-02	Authentication Latency	SHA-256 cryptographic password hashing and credential verification shall complete in < 200 ms.
NFR-PERF-03	Countdown Timer Accuracy	Per-second session countdown timer ticks shall deviate by less than ± 50 milliseconds per hour of operation.
NFR-PERF-04	PDF Compilation Throughput	Compiling and exporting a 100-row tabular PDF report via iText shall complete in < 1.5 seconds.

7.2 Reliability, Availability & Disaster Recovery

- **ACID Transaction Compliance:** All session creations, time extensions, and invoice settlements execute within strict database transaction boundaries. In the event of a runtime SQL exception or disconnection, uncommitted changes rollback automatically.
- **Database Reconnection Resilience:** The connection manager checks whether the connection is closed (`con.isClosed()`), seamlessly attempting re-establishment before throwing fatal errors.
- **Exception Logging:** All caught runtime errors are logged via `java.util.logging.Logger` for post-incident diagnostics.

7.3 Information Security & Data Protection

- **Cryptographic Password Integrity:** Passwords must never be stored, transmitted, or logged in plain text. SHA-256 hashing is enforced across all registration and change workflows.
- **SQL Injection Elimination:** Dynamic string-concatenated SQL queries are strictly prohibited. All queries with external inputs use parameterized `PreparedStatement` objects.
- **Privilege Separation:** Staff accounts are strictly prevented from viewing or modifying user accounts, default passwords, or deleting operational logs.

7.4 Software Quality Attributes

- **Usability:** Intuitive navigation with consistent iconography, informative popups (`JOptionPane`), and auto-selection of table records.
- **Maintainability:** Clean source code structure with zero inline fully-qualified package namespaces (all imports placed at top of files), modular MVC separation, and comprehensive English comments.
- **Portability:** Standard Java bytecode executes reliably across Windows, macOS, and Linux without recompilation.

8. System Verification, Validation & Test Cases

8.1 Requirements Traceability Matrix (RTM)

Requirement ID	Module / Feature Description	Test Case ID(s)	Verification Status
FR-AUTH-01	User Authentication & SHA-256 Verification	TC-AUTH-01, TC-AUTH-02	VERIFIED / PASS
FR-AUTH-02, 03	First-Time Login Mandatory Password Reset	TC-AUTH-03	VERIFIED / PASS
FR-DASH-01..03	Live Station Grid & Real-time KPI Telemetry	TC-DASH-01, TC-DASH-02	VERIFIED / PASS
FR-SES-01..03	Duration Allocation & Auto Price Calculation	TC-SES-01, TC-SES-02	VERIFIED / PASS
FR-SES-04	Dynamic in-session '+ Add Time' Extension	TC-EXT-01, TC-EXT-02	VERIFIED / PASS
FR-BILL-01..06	Dedicated BillFrame & iText PDF Receipt Export	TC-BILL-01, TC-BILL-02	VERIFIED / PASS
FR-REP-01..04	Station Defect Dispatch & Repair Resolution	TC-REP-01, TC-REP-02	VERIFIED / PASS
FR-USR-01..06	Staff User Provisioning & Credential Reset	TC-USR-01, TC-USR-02	VERIFIED / PASS

8.2 Comprehensive Test Scenarios & Verified Results

Test Case TC-AUTH-01: Valid Staff / Admin Authentication:

- **Preconditions:** Database contains active user record (username: 'admin', role: 'Admin').
- **Test Input:** Username: 'admin', Password: 'correct_password'.
- **Expected Result:** System hashes password via SHA-256, matches digest, shows 'Login Successful' dialog, records login in user_logs, and opens Dashboard.
- **Actual Result:** **PASS** - Credentials validated, shift audit log created, Dashboard opened.

Test Case TC-SES-01: Duration Selection & Real-Time Estimated Price Calculation:

- **Preconditions:** Station 'PC-01' is 'Available' with hourly rate of Rs. 200.00.
- **Test Input:** Select 'PC-01', choose duration '2 Hours (120 mins)'.
- **Expected Result:** System calculates estimated charge: $(120/60.0)*200 = \text{Rs. } 400.00$, sets timer to '02:00:00', and reveals '■ Start Session' button.
- **Actual Result:** **PASS** - Price displayed as 'Rs. 400.00', timer initialized to '02:00:00'.

Test Case TC-EXT-01: Dynamic '+ Add Time' During Ongoing Match:

- **Preconditions:** Station 'PC-01' is 'Occupied' with active session SES-001 (allocated 60 mins).
- **Test Input:** Click '■ + Add Time', select '+1 Hour (60 mins)'.
- **Expected Result:** System updates database (add_minutes = 60, total_minutes = 120, total_amount = Rs. 400.00), extends countdown timer by +3600 seconds, and refreshes live view.
- **Actual Result:** **PASS** - Additional 60 minutes added atomically, countdown timer extended.

Test Case TC-BILL-01: Session Termination & Dedicated BillFrame Invoicing:

- **Preconditions:** Station 'PC-01' session completed; player approaches checkout.
- **Test Input:** Click '■ End Session & Bill' in Dashboard, enter cash received in BillFrame, click 'Print / Save PDF Bill'.
- **Expected Result:** System terminates session, sets station status to 'Available', opens BillFrame, inserts invoice record, compiles official PDF receipt, and prompts user.
- **Actual Result:** **PASS** - BillFrame displayed, station marked available, PDF invoice exported successfully.

9. Appendices & Glossary

9.1 Glossary of Technical Terms & Acronyms

Term / Acronym	Full Definition & Context in this System
SRS	Software Requirements Specification: Formal document specifying all functional, operational, behavioral, and quality requirements.
SDLC	Software Development Life Cycle: The structured framework (Waterfall Model) governing the engineering of this system.
GUI	Graphical User Interface: Visual interactive front-end constructed via Java Swing and NetBeans Form Builder.
JDBC	Java Database Connectivity: Industry standard Java API for executing SQL queries and managing relational database transactions.
SHA-256	Secure Hash Algorithm 256-bit: Cryptographic one-way hash function used to digest and secure passwords.
FlatLaf	Flat Look and Feel: Modern, high-DPI compatible Swing look-and-feel library providing clean, professional desktop aesthetics.
NIC	National Identity Card: Government-issued citizen identity number validated using statutory regex formats.
MoSCoW	Must / Should / Could / Won't: Standard requirements prioritization framework used across all functional specifications.
RTM	Requirements Traceability Matrix: Grid mapping user requirements to system design components and verification test cases.
UAT	User Acceptance Testing: Final operational verification conducted with actual counter receptionists and administrative staff.
POS	Point of Sale: Counter hardware ecosystem including thermal receipt printers and barcode/QR scanners.

9.2 Future Strategic Enhancements & Web/Cloud Roadmap

While the present release satisfies all baseline desktop operational requirements, the system architecture was engineered to accommodate future modular extensions:

- **Cloud Synchronized Multi-Branch Database:** Migration of the local MySQL instance to a managed cloud database (e.g., AWS RDS / Google Cloud SQL) for multi-branch esports franchises.
- **Web & Mobile Online Seat Reservation Portal:** Lightweight web frontend enabling gamers to check station availability and book gaming slots remotely.
- **Online Cashless Payment Gateway Integration:** LankaQR / Stripe / IPG integration for processing card and mobile wallet payments upon QR token scan.
- **Steam / Epic Games Platform API Integration:** Automated client-side background service to sync player tournament statistics and save games upon login.

--- END OF SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT ---

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